

Unlocking Behavioral Intelligence in Airline Loyalty Programs

Churn prediction, customer value segmentation, and a retention playbook for 16,737 Canadian loyalty members (2012–2018)

Business Analytics Team · Technical Report · Companion artifacts: [Airline_Loyalty_Churn_Colab.ipynb](#) (reproducible end-to-end), [retention_dashboard.html](#) (working prototype)

1. Executive Summary

We built a leakage-safe churn model, a behavioral segmentation, and a retention playbook on 24 months of flight activity for the live member base. Validated strictly out-of-time, the model concentrates **55% of all future churners into the riskiest 10% of members (5.5× lift)**. Applied to the December 2018 base of 14,670 active members, it identifies **1,228 high-risk members carrying \$9.7M of recorded customer lifetime value**, of whom 619 are high-value enough to justify personal outreach (P1). Three findings should change how the program is run: silence — not cancellation — is the real churn signal and it is visible months in advance; members who have ever redeemed points churn at 2.6% versus 11.5% for never-redeemers; and the CLV metric the business tracks is backward-looking enough to actively mislead targeting.

2. Problem Framing

2.1 What counts as churn

Formal cancellations are too rare and too late to manage the program: only 2,067 of 16,737 members ever cancel, and in our test window 322 members cancelled while still flying, while 439 others simply went silent without cancelling. We therefore define churn behaviorally: **a member has churned if, in the six months following a snapshot date, they either formally cancel or book zero flights**. Six months matches the marketing planning cycle: a one-month horizon predicts the unpreventable, while a twelve-month horizon would consume half of the 24-month activity window in labels alone.

2.2 No future information — the snapshot design

Every feature is computed strictly from data on or before a snapshot date; every label strictly after it. Validation is **out-of-time**: train on the December 2017 snapshot (features Jan–Dec 2017, outcomes Jan–Jun 2018), test on the June 2018 snapshot (outcomes Jul–Dec 2018), and score the live base at December 2018 for the dashboard. A random row split was rejected because it would let the same member appear on both sides and would share seasonality between train and test — the two most common leakage paths in churn modeling.

Snapshot	Features from	Labels from	Role	n	Churn rate
Dec 2017	Jan–Dec 2017	Jan–Jun 2018	Train	12,305	5.0%
Jun 2018	Jan 2017–Jun 2018	Jul–Dec 2018	Out-of-time test	13,699	5.6%
Dec 2018	Full 24 months	— (unknown future)	Production scoring	14,670	—

3. Data Audit and Cleaning Decisions

The data is real-world in character, and the audit surfaced four substantive issues. Each fix is documented with the rejected alternative, because cleaning choices are modeling choices.

Issue	Magnitude	Decision	Why (and what was rejected)
Exact duplicate activity rows	1,922 rows	Drop	Identical on all 8 columns → double-loaded records. Keeping them double-counts flights.
Conflicting duplicates (same member-month, different values)	1,949 rows	Sum within member-month	Differing values resemble split booking postings. Averaging understates activity; dropping deletes real flights.
Negative salaries	20 members	Absolute value	Magnitudes are plausible; sign-entry error is the likeliest cause. Nulling discards usable signal.
Salary missing for 100% of College members	4,238 (25% of base)	Median impute + salary_missing flag	Structurally missing (MNAR): missingness IS the College indicator. The flag preserves that information; dropping 25% of members would bias every downstream estimate.

Two structural facts shaped the design: flight activity exists only for Jan 2017–Dec 2018 even though enrollments date to 2012 (tenure is therefore a profile feature, not an activity feature), and 34,302 zero-flight rows appear after members' cancellation dates — calendar padding confirming the activity table is a dense member × month grid, not an event log. We also found 7 members who redeemed points despite zero recorded flights in 24 months (“**phantom redeemers**”); they are flagged for a finance/IT integrity audit and excluded from marketing segments.

4. Churn Model

4.1 Features and models

Fifteen features were engineered per snapshot: recency of last flight, active months, totals for flights/distance/points, redemption rate and dollars, **momentum** (flights in the last 6 months versus the member's own prior 6 months — designed to catch decline before silence), average distance per flight, tenure, salary with missingness flag, card tier, education, marital status, gender, and the 2018-promotion flag. CLV was deliberately included to test whether historical value predicts future behavior. Two models were trained with class weighting (5–6% positive rate): a logistic-regression baseline and a histogram gradient-boosting classifier.

4.2 Results — judged on the metric that matches the decision

At a 5.6% base rate, ROC-AUC flatters any model; the campaign question is “if we can only contact N members, how many churners do we catch?”, which is precision-recall territory. Gradient boosting wins decisively where it matters: **PR-AUC 0.51 versus 0.26** for the baseline (≈9× better than random targeting), with ROC-AUC 0.79. Top-decile lift is 5.5×: contacting the riskiest 10% of members reaches 55% of all actual churners.

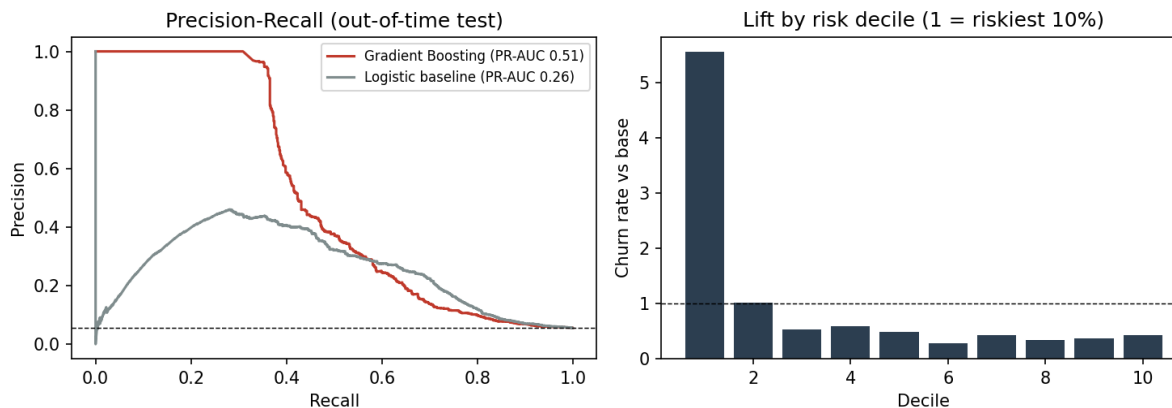


Figure 1 — Out-of-time test performance: precision-recall curves (left) and lift by risk decile (right).

4.3 What drives churn

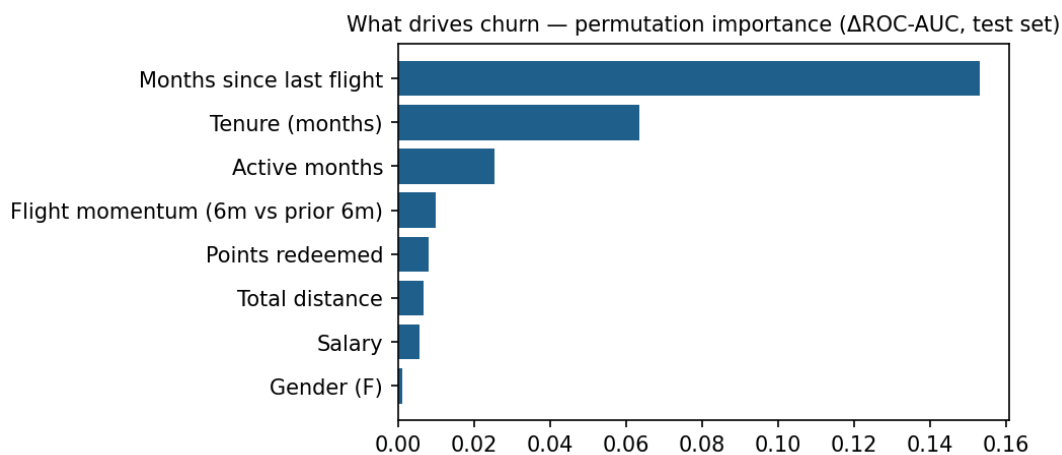


Figure 2 — Permutation importance on the held-out test set (Δ ROC-AUC).

- **Recency dominates:** months since last flight is worth more than the next five features combined. Silence is the alarm, and it rings months before any cancellation form is filed.
- **Tenure is second:** the first ≈ 12 months of membership are a cliff; new members churn at multiples of veterans' rates.
- **Momentum catches decline in motion:** members still flying, but below their own prior baseline.
- **CLV ranks 21st of 24 features.** Historical spend barely predicts future behavior — direct evidence for Section 5's critique.
- **Demographics are nearly irrelevant once behavior is known.** Gender, education and marital status add almost nothing: target behavior, not personas.

5. Segmentation and the CLV Critique

5.1 Why a hybrid design

Pure K-Means over all members produced statistically separable but unactionable output: a 7-member “cluster” (the phantom-redeemer anomaly) and 612 never-flyers blended into active groups. We therefore segment in two stages: rules first (Phantom Redeemers → audit; zero flights in 24 months → Sleeping Enrollees), then K-Means on the remaining 14,051 flyers over six behavioral features (log-scaled where skewed, outliers clipped). Silhouette analysis put $k=3$ and $k=4$ in a statistical tie (0.321 vs 0.325); **$k=3$ was chosen because the fourth cluster held only 12 members** — distinctness without actionability is noise.

Segment	n	Avg churn risk	Avg CLV	Profile	Play
Core Frequent Flyers	10,631	18%	\$7,952	39 flights/24m, 14 active months, long tenure	Protect, don't discount. No retention spend.
Cooling Veterans	1,830	24%	\$7,828	Long tenure, negative momentum, ≈3 months silent	“We’ve missed you” + tier-protection notice at day 60 of silence
New Climbers	1,590	50%	\$8,177	8-month tenure, high momentum, 52% high-risk	First-redemption onboarding journey in first 90 days
Sleeping Enrollees	612	58%	\$8,148	Zero flights in entire 24-month window	One points-expiry win-back, then suppress from paid campaigns
Phantom Redeemers	7	—	\$5,766	Redeem points with zero recorded flights	Finance/IT data-integrity audit

5.2 The number that indicts the CLV column

Sleeping Enrollees have not flown in at least 24 months, yet their average recorded CLV is **\$8,148 — above the base-wide average of \$7,989**. A manager ranking a report by CLV would actively prioritize members who are already gone. The metric answers “who was valuable,” never “who will be”: it is total historical invoice value, static, and (per Section 4.3) nearly uninformative about future behavior. We propose the program adopt **Forward Value = CLV × (1 – churn risk)** as the default ranking in all retention reporting; it is computed for every member in the prototype.

6. Retention Playbook — who, when, why, in what form

Risk alone does not tell marketing what to do. Crossing risk bands (High >60% / Medium / Low <30%) with value bands (CLV top 20% / mid 30% / standard 50%) sets the **priority and budget**; the segment sets the **content** of the intervention.

Priority	Who (Dec 2018)	Trigger & timing	Intervention (hand-to-ops form)
P1	619 members: high risk × top/mid value	Within 7 days of entering the band; re-scored monthly	Loyalty-desk personal call + redemption-unlock offer: “your points now cover a Toronto–Vancouver return — book by [date]”. Rationale: ever-redeemers churn at 2.6% vs 11.5% for never-redeemers — the strongest retention behavior in the data.

P2	892 members: high risk $\times$ standard value, or medium risk $\times$ top value	Day 60 of inactivity (recency = 2 months is the empirical inflection)	Automated lifecycle email: point balance + three one-click redemption suggestions priced below the member's current balance. Sleeping Enrollees instead get a single points-expiry notice, then suppression.
P3	13,159 members: everyone else	Quarterly	Statement only. The model's job is equally to tell marketing who NOT to pay to retain — 90% of the base requires no retention spend.

Segment overlays sharpen the message: New Climbers receive the first-90-days redemption journey (their problem is an onboarding cliff, not disengagement); Cooling Veterans receive the tier-protection framing, because the data shows tier alone does not retain (test-window churn is 5.3% Star, 5.6% Nova, 6.1% Aurora — flat to rising up the ladder).

7. Working Prototype

The deliverable is **retention_dashboard.html**: a single self-contained file (no install, no server, no manual) holding the full scored December 2018 base. It opens to five KPIs, clickable segment cards with their plays, a Risk $\times$ Value matrix, and an action list pre-sorted with the most urgent members on top. Each row carries the member number, segment, priority, churn risk, historical CLV, forward value, months silent, and the specific recommended action; filters cover priority, segment, province and member search, and one button exports the filtered call list as CSV for the operations team. A Streamlit version (app.py) is provided for teams preferring a hosted tool, and the Colab notebook regenerates all inputs from the four raw CSVs with a fixed seed.

8. Three Recommendations a CFO and a CMO Can Both Sign

1. **Launch a first-redemption program and measure it as a retention lever.** Ever-redeemers churn at 2.6% versus 11.5%. Waive the redemption minimum once in each member's first 90 days and run it as an A/B test with a holdout. CFO case: redemption liability is spent points, cheaper than re-acquisition; CMO case: it converts the highest-risk segment (New Climbers, 52% high-risk) at its most reachable moment. Even a one-point churn reduction among the 1,590 New Climbers protects roughly \$130K of CLV per cycle.
2. **Replace CLV with Forward Value (CLV $\times$ survival probability) in all retention reporting and budget allocation.** CFO case: \$9.7M of "value" currently on the books sits in the high-risk band and is being reported as if secure; CMO case: stops campaigns from chasing high-CLV members who are already gone (612 Sleeping Enrollees average \$8,148 CLV and zero flights in two years).
3. **Re-aim retention spend by behavior, not tier, using the P1/P2/P3 bands.** Churn barely varies by card tier, so tier-based perks are loyalty theater for retention purposes. Concentrate the human-touch budget on the 619 P1 members (4% of the base) and automate P2; release the spend currently spread across the 13,159 P3 members. CFO case: smaller, measurable budget; CMO case: 5.5 $\times$ lift means each contacted member is 5–6 times more likely to be a genuine save opportunity.

9. Limitations and Next Steps

- 24 months of activity permits one full out-of-time cycle; with longer history, rolling-origin validation across many snapshots would tighten error bars.
- Scores are rank-reliable, not calibrated probabilities — add isotonic calibration before using them in expected-value arithmetic.
- The redemption–retention effect is correlational and selection-confounded; the playbook launches as A/B tests with holdouts, converting the recommendations into measured incremental revenue.
- The 2018 Promotion cohort (971 members, 11 months of data) scores low-risk but has not faced a renewal cycle; re-score quarterly.

Reproducibility: every number in this report regenerates from the four raw CSVs by running the companion notebook top to bottom (seed = 42).